Lecture 1

1. **The juxta-glomerular apparatus is formed from three cells:**
   a) Juxta-glomerular cells, macula densa and lacis cells.
   b) Macula densa, mesangial cells and lacis cells.
   c) Juxta-medullary cells, podocytes, and macula densa.
   d) Macula densa, Bowman's capsule and lacis cells.

Answer: A

2. **Angiotensin II causes:**
   a) Stimulation of RBCs production
   b) Increase of peripheral resistance.
   c) Formation of vitamin D
   d) Control of calcium excretion.

Answer: B

3. **Sympathetic stimulation of the kidneys:**
   a) Decreases the GFR
   b) Increases the GFR
   c) No change in GFR
   d) Increases urine formation

Answer: A

4. **Increases in arterial pressure above 180 mmHg:**
   a) Increases GFR
   b) Decreases GFR
   c) No change in GFR
   d) First decreases then increase GFR

Answer: A

5. **Autoregulation depends upon the concentration at macula densa of:**
   a) Calcium chloride.
   b) Sodium chloride.
   c) Potassium chloride.
   d) Magnesium chloride.

Answer: B

6. **How does the juxtaglomerular apparatus respond when systemic blood pressure increases?**

a) The juxtaglomerular cells send a message to the afferent arteriole to dilate.
b) The macula densa sends a message to the efferent arteriole to constrict.
c) The macula densa sends a message to the afferent arteriole to constrict.
d) The granular cells release renin which causes systemic arterioles to constrict.

Answer: C

7. **Four sections of the vasculature of the kidney tubule are listed below. Which one lists them in correct order of blood flow from left to right?**
   a) Efferent arteriole, glomerulus, afferent arteriole, peritubular capillaries
   b) Afferent arteriole, glomerulus, efferent arteriole, peritubular capillaries
   c) Peritubular capillaries, afferent arteriole, glomerulus, efferent arteriole
   d) Glomerulus, afferent arteriole, peritubular capillaries, efferent venule

   Answer: B

8. **Complete the following sentence correctly. Angiotensin II:**
   a) Stimulates the adrenal glands to release aldosterone
   b) Increases potassium reabsorption from the filtrate
   c) Increases sodium (Na^+) excretion at the kidneys
   d) Reduces our thirst

   Answer: A

9. **Macula densa is present in:**
   a) Collecting tubule
   b) Proximal convoluted tubule
   c) Distal convoluted tubule
   d) Afferent arteriole

   Answer: C

10. **The glomerulus:**
    a) Has both afferent and efferent arterioles
    b) Contains capillaries that are at a higher hydrostatic pressure than the peritubular capillaries
    c) Contains renin-secreting cells in its afferent arteriole.
    d) is lined by fenestrated endothelium
    e) all of the above.

    Answer: E

11. The juxtamedullary nephrons are characterized by:

a) Their glomeruli lie in the inner part of the cortex near the medulla.
b) Represent about 85% of the total number of nephrons

c) Their loops of Henle are short and penetrate to short distance in the medulla
d) Surrounded by U-shaped capillaries called the peritubular capillaries.

Answer: A

12. The juxtaglomerular apparatus secretes:

a) Renin.
b) Erythropoietin.
c) Vitamin D.
d) Angiotensin II.

Answer: A

13. **The cortical nephrons when compared to the juxtamedullary nephrons cortical nephrons have:**
    a) longer loops of Henle.
    b) greater numbers.
    c) vasa recta.
    d) a major role in urine concentration.
    e) lower blood supply.

    Answer: B

14. **The juxtaglomerular cells of the JGA secrete**:
    a) aldosterone.
    b) ADH.
    c) angiotensin II.
    d) renin.
    e) adrenaline.

    Answer: D

15. **Concerning the juxtaglomerular (JGA) apparatus which of the following is correct:**
    a) it is located in the area of contact between the afferent arteriole, efferent arteriole, and DCT.
    b) JG cells act as receptors for NaCl concentration.
    c) macula densa cells secrete rennin.
    d) its main function is to regulate glucose concentration in the plasma.
    e) it is present in the juxtamedullary nephrons only.

    Answer: A

16. **Macula densa:**
    a) is present in the collecting ducts.
    b) secretes renin.
    c) responding to an increase in glucose conc.

d) detects NaCl load.
e) is present in media of afferent arteriole.

Answer: D

17. **Which of the following is not a function of the kidney:**
a) regulation of solutes concentration.
b) production of renin.
c) production of angiotensin.
d) activation of vitamin D.
e) urine formation.

Answer: C

18. **All of the following are functions of the kidney EXCEPT:**
a) regulation of acid-base balance.
b) production and release of renin.
c) conversion of angiotensin I into angiotensin II.
d) activation of vitamin D.
e) regulation of the volume and ionic composition of body fluids.

Answer: C

19. Renin hormone is secreted:
a) from juxtaglomerular cells.
b) from macula densa cells.
c) from epithelial cells.
d) in response to increased blood flow to the kidney.
e) in response to increased NaCl.

Answer: A

20. **The functions of the kidney include:**
a) regulation of blood pressure.
b) regulation of the volume, composition, and pH of body fluids.
c) control of the rate of RBCs production.
d) all of the above are correct.
e) only b & c are correct.

Answer: D